

7.1 Suppose that we decompose the schema $R = (A, B, C, D, E)$ into

$$\begin{aligned} &(A, B, C) \\ &(A, D, E). \end{aligned}$$

Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds:

$$\begin{aligned} &A \rightarrow BC \\ &CD \rightarrow E \\ &B \rightarrow D \\ &E \rightarrow A \end{aligned}$$

1. **Compute $R_1 \cap R_2$:**
 - $R_1 = (A, B, C), R_2 = (A, D, E)$
 - $R_1 \cap R_2 = \{A\}$.
2. **Check if A is a superkey:**
 - Given $A \rightarrow BC, A^+ = \{A, B, C\}$.
 - While A is not a superkey for R , it **is a superkey for R_1** .
3. **Lossless Condition:**
 - Since $R_1 \cap R_2 = \{A\} \rightarrow \{A, B, C\} = R_1$, the decomposition is lossless.

Final Answer:

The decomposition is lossless because $R_1 \cap R_2 = \{A\}$ and $A \rightarrow BC$ makes A a superkey for R_1 .

7.13 Show that the decomposition in Exercise 7.1 is not a dependency-preserving decomposition.

1. **Original Dependencies (F):**

$$F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$$

2. **Decomposed Dependencies:**
 - $F_1 = \{A \rightarrow BC\}$ (in $R_1 = (A, B, C)$).
 - $F_2 = \{E \rightarrow A\}$ (in $R_2 = (A, D, E)$).
3. **Lost Dependencies:**

- $CD \rightarrow E$: CD and E are split across R_1 and R_2 .
- $B \rightarrow D$: B and D are split across R_1 and R_2 .

Conclusion:

The decomposition is **not dependency-preserving** because $CD \rightarrow E$ and $B \rightarrow D$ cannot be derived from $F_1 \cup F_2$.

Final Answer:

The decomposition is not dependency-preserving because $CD \rightarrow E$ and $B \rightarrow D$ cannot be derived from $F_1 \cup F_2$.

7.21 Give a lossless decomposition into BCNF of schema R of Exercise 7.1.

1. **Identify BCNF Violations:**

- $B \rightarrow D$: B is not a superkey.

2. **Decompose:**

- Split R into (B, D) and (A, B, C, E) .
- Verify (A, B, C, E) :
 - $E \rightarrow A$: E is a superkey ($E^+ = R$).
 - $A \rightarrow BC$: A is not a super key
- Split (A, B, C, E) into (A, B, C) and (A, E)
 - $R_2 = (A, B, C)$
 - $A \rightarrow BC$: A is a super key
 - $R_3 = (A, E)$
 - $E \rightarrow A$: E is a super key

Final BCNF Decomposition:

$$\begin{aligned} R_1(B, D), \quad F_1 &= \{B \rightarrow D\} \\ R_2(A, B, C, E), \quad F_2 &= \{A \rightarrow BC\} \\ R_3(A, E), \quad F_3 &= \{E \rightarrow A\} \end{aligned}$$

7.22 Give a lossless, dependency-preserving decomposition into 3NF of schema R of Exercise 7.1.

Final Answer:

$$\begin{array}{l} R_1(A, B, C), \quad F_1 = \{A \rightarrow BC\} \\ R_2(C, D, E), \quad F_2 = \{CD \rightarrow E\} \\ R_3(B, D), \quad F_3 = \{B \rightarrow D\} \\ R_4(A, E), \quad F_4 = \{E \rightarrow A\} \end{array}$$

7.30 Consider the following set F of functional dependencies on the relation schema (A, B, C, D, E, G) :

$$\begin{array}{l} A \rightarrow BCD \\ BC \rightarrow DE \\ B \rightarrow D \\ D \rightarrow A \end{array}$$

- Compute B^+ .
- Prove (using Armstrong's axioms) that AG is a superkey.
- Compute a canonical cover for this set of functional dependencies F ; give each step of your derivation with an explanation.
- Give a 3NF decomposition of the given schema based on a canonical cover.
- Give a BCNF decomposition of the given schema using the original set F of functional dependencies.

a. $B^+ = ABCDE$

$$(b) \quad \because A \subseteq AG \quad G \subseteq AG \quad \therefore AG \rightarrow A \quad AG \rightarrow G$$

$$\therefore B \subseteq BCD \quad C \subseteq BCD \quad D \subseteq BCD$$

$$\therefore BCD \rightarrow B \quad BCD \rightarrow C \quad BCD \rightarrow D$$

according to Transitivity

$$\begin{cases} A \rightarrow BCD \\ BCD \rightarrow B \end{cases} \Rightarrow A \rightarrow B \quad \text{with the same method.}$$

we can conclude $A \rightarrow C \quad A \rightarrow D \quad A \rightarrow BC$

$$\begin{cases} A \rightarrow BC \\ BC \rightarrow DE \\ E \subseteq DE \end{cases} \Rightarrow A \rightarrow E$$

$$\therefore A \subseteq AG \quad \therefore AG \rightarrow A \quad \therefore AG \rightarrow B \quad AG \rightarrow C \quad AG \rightarrow D \quad AG \rightarrow E$$

We can conclude that $(AG)^+ = ABCDE$.

So, AG is a superkey

(c) ① use decomposition

$$F = \{A \rightarrow B, A \rightarrow C, A \rightarrow D, BC \rightarrow D, BC \rightarrow E, B \rightarrow D, D \rightarrow A\}$$

② try to delete duplicate attributes

consider that $BC \rightarrow D$ and $BC \rightarrow E$ $\begin{cases} \text{if delete } B, (C)^+ = C, \text{ cannot} \\ \text{if delete } C, (B)^+ = ABCDE, \text{ can} \end{cases}$

$$\therefore F = \{A \rightarrow B, A \rightarrow C, A \rightarrow D, B \rightarrow D, B \rightarrow E, D \rightarrow A\}$$

③ try to delete duplicate function dependencies.

$\therefore A \rightarrow B, B \rightarrow D \therefore A \rightarrow D$ is a duplicate fd.

$$\therefore F = \{A \rightarrow B, A \rightarrow C, B \rightarrow D, B \rightarrow E, D \rightarrow A\}$$

$$(d) \quad R_1 = \{A B C\} \quad R_2 = \{B D E\} \quad R_3 = \{D A\}$$

$$(e) \quad F = \{A \rightarrow BCD, BC \rightarrow DE, B \rightarrow D, D \rightarrow A\}$$

We can conclude that candidate keys are A, B, D

$$① \quad R_1 = \{AD\}, \quad \text{remained } R_2 = \{BCDE\}$$

$$② \quad \text{for } R = (BCDE) \quad F = \{BC \rightarrow DE, B \rightarrow D\}$$

candidate key: BC $\therefore BC \rightarrow DE$ satisfies the BCNF.

$$\therefore R_2 = (BCDE)$$

$$\text{All in all, } R_1 = (AD) \quad R_2 = (BCDE)$$

7.32 Consider the schema $R = (A, B, C, D, E, G)$ and the set F of functional dependencies:

$$\begin{aligned}A &\rightarrow BC \\ BD &\rightarrow E \\ CD &\rightarrow AB\end{aligned}$$

- Find a nontrivial functional dependency containing no extraneous attributes that is logically implied by the above three dependencies and explain how you found it.
- Use the BCNF decomposition algorithm to find a BCNF decomposition of R . Start with $A \rightarrow BC$. Explain your steps.
- For your decomposition, state whether it is lossless and explain why.
- For your decomposition, state whether it is dependency preserving and explain why.

a. We can calculate the closure of the left attributes:

$$(A)^+ = ABC$$

$$(BD)^+ = BDE$$

$$(CD)^+ = ABCDE$$

$\therefore$ we can find a new nontrivial functional dependency $CD \rightarrow E$

b. If we start with $A \rightarrow BC$, then we can take a decomposition

$$R_1 = (ABC), R_2 = (ADEG)$$

Because there is no effective functional dependency from R_2 , it satisfies the BCNF

solution:

$\begin{aligned}R_1(A, B, C), \quad F_1 &= \{A \rightarrow BC\} \\ R_2(A, D, E, G), \quad F_2 &= \emptyset\end{aligned}$
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c. $\because R_1 \cap R_2 = A$, A is the superkey of R_1 , so the decomposition is lossless.

d. We should check the original functional dependencies of R .

- $A \rightarrow BC$ is in F_1
- $BD \rightarrow E$: $\because B$ is not in the same decomposition as D, E , $\therefore$ it cannot be dependency preserving.

solution

The decomposition is not a dependency – preserving decomposition.